STARTER MOTOR (Continued)

STARTER MOTOR TESTING

(1) Remove starter motor from vehicle. Refer to Starter Motor Removal / Installation.

(2) Mount starter motor securely in a soft-jawed bench vise. The vise jaws should be clamped on mounting flange of starter motor. Never clamp on starter motor by field frame.

(3) Connect a suitable volt-ampere tester and12-volt battery to starter motor in series, and set ammeter to 100 ampere scale. See instructions provided by manufacturer of volt-ampere tester being used.

(4) Install a jumper wire from solenoid terminal to solenoid B(+) terminal stud. The starter motor should operate. If starter motor fails to operate, replace faulty starter motor.

(5) Adjust carbon pile load of tester to obtain free running test voltage. Refer to starter motor free running test voltage specifications.

(6) Note reading on ammeter and compare reading to free running test maximum amperage draw. Refer to starter motor free running test maximum amperage draw specifications.

(7) If ammeter reading exceeds maximum amperage draw specification, replace faulty starter motor.

STARTER SOLENOID TESTING

This test can only be performed with starter motor removed from vehicle.

(1) Remove starter motor from vehicle. Refer to **Starter Motor** for removal/installation.

(2) Disconnect wire from solenoid field coil terminal.

(3) Check for continuity between solenoid terminal and solenoid field coil terminal with a continuity tester (Fig. 7). There should be continuity. If OK, go to Step 4. If not OK, replace faulty starter motor.

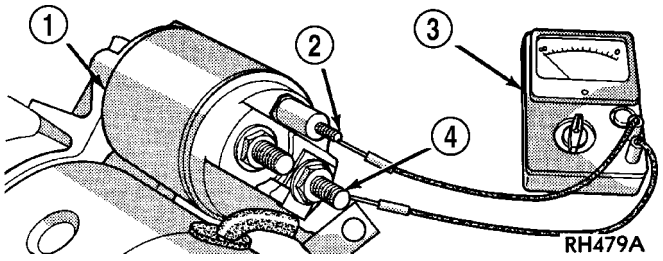


Fig. 7 CONTINUITY TEST BETWEEN SOLENOID TERMINAL AND FIELD COIL TERMINAL

1 - SOLENOID
2 - SOLENOID TERMINAL
3 - OHMMETER
4 - FIELD COIL TERMINAL

(4) Check for continuity between solenoid terminal and solenoid case (Fig. 8). There should be continuity. If not OK, replace faulty starter motor.

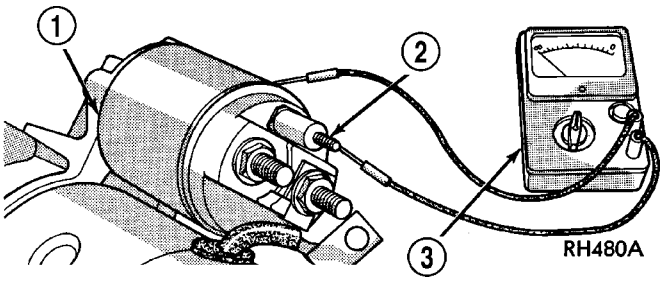


Fig. 8 CONTINUITY TEST BETWEEN SOLENOID TERMINAL AND SOLENOID CASE - TYPICAL

1 - SOLENOID
2 - SOLENOID TERMINAL
3 - OHMMETER

REMOVAL

2.4L

(1) Disconnect and isolate negative battery cable.

(2) Raise and support vehicle.

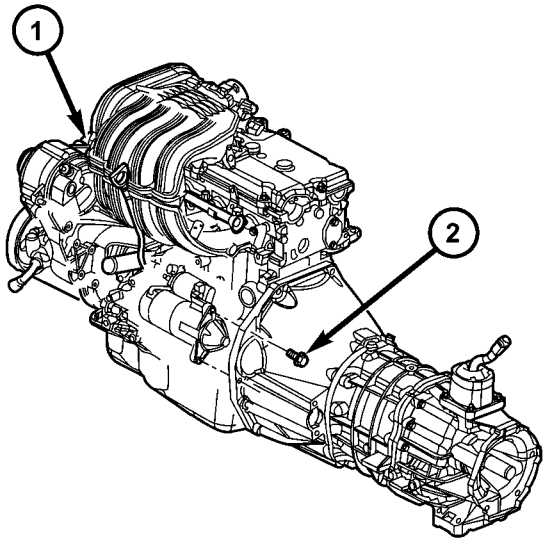
(3) While supporting starter motor, remove two bolts securing starter motor to transmission (Fig. 9).

(4) Lower starter motor far enough to access and remove nut securing battery cable to starter solenoid B(+) terminal stud (Fig. 10). Always support starter motor during this process. Do not let starter motor hang from wire harness.

(5) Remove battery cable at starter.

(6) Disconnect solenoid terminal wire harness connector from starter solenoid.

(7) Remove starter motor.



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Fig. 9 STARTER - 2.4L

1 - STARTER
2 - MOUNTING BOLTS (2)